
Identity-Preserving Neural Style Transfer for Personalized Children's Book Illustration (Computer Vision)

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Abstract

Standard Neural Style Transfer (NST) fails to preserve facial identity, hindering effective personalization for storybook illustration. We propose an Identity-Guided Style Transfer network, augmenting a fast feed-forward architecture with a novel Identity-Preserving Loss $\mathcal{L}_{identity}$ based on deep facial embedding similarity to maintain subject recognition. This method is applied to synthetic child face datasets and evaluated against baselines using metrics for identity preservation, style fidelity, and content fidelity.

1 Problem and Motivation

Problem: Traditional Neural Style Transfer (NST) struggles to preserve fine-grained facial identity when transferring a new style, often resulting in a stylized image where the content person is unrecognizable. Creating a consistently high-quality, recognizable, and stylized portrait of a child for personalized book illustration is a labor-intensive process that requires artistic skill.

Why It's Interesting: This project enables personalized storytelling, making reading far more engaging for children. It solves a real-world use case that integrates art, personalization, and deep learning. In addition, it demonstrates how deep learning can be applied to creative domains, making it appealing to both technical and general audiences.



Figure 1: Example of NST transforming a child's portrait into an artistic illustration [1].

2 Related Work

Neural Style Transfer (NST) began with the iterative optimization method [2] and progressed to fast, feed-forward architectures like AdaIN [3]. However, preserving specific content identity remains a major challenge in stylization. Our approach builds on identity regularization techniques, similar to

those used in [4], to explicitly prioritize subject recognition. We also consider leveraging advanced architectures, such as Vision Transformers [5], for improved structural consistency as a stretch goal.

3 Challenges

- **Identity vs. Style Trade-off:** maintaining recognizability under strong stylization.
- **Cross-Style Generalization:** achieving consistency across different artistic domains.
- **Compute Efficiency:** limited GPU resources for training or fine-tuning advanced Style Transfer models.

4 Dataset

The project requires two distinct sets of images: content (child portraits) and style (artistic illustrations). For **content images**, we will use the Children’s Face Dataset (from StyleGAN), which provides 10,000 AI-generated images of children’s faces. This dataset is critical for ethical compliance, as using synthetic, non-real faces entirely removes all privacy and legal concerns related to using photographs of real children. For **style images**, we will use the Hugging Face 6chan/children-hand-drawn-style-transfer dataset. This collection is specifically curated for children’s style transfer projects and includes a high diversity of hand-drawn styles such as oil paintings, colored pencil drawings, crayon drawings, and various sketches. This diversity will allow us to thoroughly test the generalization and consistency of our proposed identity-preserving method. Both datasets are **publicly available**.

5 Method and Algorithm

We will adopt an algorithm/method approach focused on improving facial identity preservation in style transfer.

Core Goal: Identity-Preserving Style Transfer

- **Baseline:** Implement a Perceptual Loss-based Fast Style Transfer Model (e.g., using a WCT or AdaIN architecture).
- **Proposed Novel Method:** Modify the transfer network by adding an Identity-Preserving Loss Function $\mathcal{L}_{identity}$ to the total loss.

$$\mathcal{L}_{total} = \alpha\mathcal{L}_{content} + \beta\mathcal{L}_{style} + \gamma\mathcal{L}_{identity}.$$

- **Identity Loss:** $\mathcal{L}_{identity}$ will be calculated as $1 - g(similarity)$, where $g(similarity) = \text{cosine_similarity}(f(Content), f(Stylized))$ and $f(\cdot)$ is the embedding from a pretrained CNN-based Face Recognition Network, used to guarantee that the output face remains recognizable.

Stretch Goal: Vision Transformer (ViT) Backbone If time and resources allow, we will investigate replacing the traditional VGG-based perceptual loss backbone with a pretrained Vision Transformer (ViT) that is slightly finetuned. The aim is to take advantage of ViT’s superior global attention mechanisms, similar to recent transformer-based style transfer architectures [5], to produce better identity consistency across the entire facial structure.

6 Evaluation

Our evaluation will be qualitative and quantitative. **Qualitatively**, we will generate side-by-side plots for multiple synthetic child faces, showing the results from both the Baseline and the Proposed Method across various artistic styles (e.g., Crayon, Oil Painting). The goal is to visually demonstrate that our Proposed Method maintains the unique facial features of the child more effectively. **Quantitatively**, we will use three metrics for comparison against the baseline: Identity Preservation (Cosine Similarity of Face Embeddings; Higher is better), Style Fidelity (Style Loss; Lower is better), and Content Fidelity (Content Loss; Lower is better).

References

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